A variable-k Turan-density lower bound for augmentations of a complete hypergraph

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Editorial note. Complete original author text and supporting proofs retained; original AI-assistance discussion and acknowledgement kept visible and recorded. No proof repair, added assumptions, AI-free-authorship claim or independent correctness certification.

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Allowed standard prerequisites (editorial).

- Hypergraph Turan densities, finite fields and exponential maximal inequalities. Native augmentation and zero-kernel definitions, random-map constructions, lifting arguments and complete cyclic-scan estimates retained.. All selected construction/iteration/maximal-bound obligations remain present as original author arguments.

1 Introduction

An *r-graph* H with vertex set V is a collection of r-subsets of V. The *Turán number* of an r-graph H, denoted by $\mathrm{ex}(n,H)$, is the maximum number of edges in an H-free r-graph on n vertices. As it was shown by Katona, Nemetz and Simonovits [33] (see Lemma 5.4), the limit

$$\pi(H) := \lim_{n\to\infty} \mathrm{ex}(n,H)/\binom{n}{r},$$

exists; we call it the *Turán density of H*.

Since the fundamental paper by Turán [53], Turán problems have been a central topic in extremal combinatorics. For *graphs* (that is, 2-graphs), Turán densities are well understood: the classical Erdős–Stone–Simonovits theorem [19, 18] determines $\pi(H)$ for any given graph H in terms of the chromatic number of H. However, determining the Turán density of a given r-graph with $r \ge 3$ is very difficult in general; we refer the reader to surveys [24, 15, 48, 34].

In order to gain insights, considerable efforts were put towards understanding the growth of $\pi(F_r)$ as $r \to \infty$ for some natural sequences of forbidden r-graphs F_r.

One such (2-parameter) forbidden family is as follows. For $2 \le k \le r+1$, let H^r_k be the (unique up to an isomorphism) r-graph on $r+1$ vertices with k edges. It is easy to see that $\pi(H^r_2) = 0$, so we consider $k \ge 3$ only. While an easy averaging argument shows $\pi(H^r_k) \le (k-1)/(r+1)$, a slightly better bound

$$\pi(H^r_k) \le \frac{k-2}{r}, \quad \text{for every } r+1 \ge k \ge 3 \tag{1.1}$$

follows from Sidorenko [51, Eq. (7)]; for an easily accessible paper containing a proof of (1.1), see e.g. [40, Corollary 3.9].

Various special cases of the Turán problem for H^r_k are of great interest in hypergraph Turán theory. For example, if $k = r+1$ then H^r_k is K^r_{r+1}, where K^r_m denotes the complete r-graph on m vertices. While the old theorem of Mantel [38] shows that the Turán density of the triangle K^2_3 is $1/2$, no density $\pi(K^r_{r+1})$ with $r \ge 3$ is known. In particular, the famous Tetrahedron Conjecture that $\pi(K^3_4) = 5/9$ is still open, despite a \$1000 reward offered by Erdős (see [24, Page 260]). The currently

best known upper bound $\pi(K_4^3) \le 0.5577...$ was obtained by Jeong, Park, Im, Lee and Yang [32] using flag algebras. In the next case $r = 4$, a construction of Giraud [26] gives $\pi(K_5^4) \ge 11/16$, which is conjectured to be sharp. Here, the best known upper bound $\pi(K_5^4) \le 0.7365...$ is due to Markström [39]. Interestingly, Sidorenko [48] in his survey mentioned that there was no plausible conjecture about the value of $\pi(K_{r+1}^r)$ for any $r \ge 5$ and this still seems to be the case today. However, the behaviour of $\pi(K_{r+1}^r)$ as $r \to \infty$, studied in e.g. [51, 14, 35, 52, 22, 49, 11, 37, 42, 25] is now fairly well understood: we know that

$$1 - \frac{4}{r+2} \le \pi(K_{r+1}^r) \le 1 - \frac{1}{r}, \quad \text{for every } r \ge 2.$$

See [43, 25] for connections of this problem to covering codes.

Another notable case is H_3^r (that is, the r-graph with $r+1$ vertices and 3 edges). If $r = 3$ then we get K_4^3 minus an edge, for which Frankl and Füredi [21] (and independently Giraud, see [21, Page 325]) showed that $\pi(H_3^3) \ge 2/7$, which is conjectured to be sharp (with the best known upper bound $\pi(H_3^3) \le 0.2868...$ obtained by Falgas-Ravry and Vaughan [20] using flag algebras). Gunderson and Semeraro [28] proved that $\pi(H_3^4) = 1/4$. Sidorenko [50] presented a geometric construction of H_k^r-free r-graphs for arbitrary r and k; in particular, it gives quite good lower bounds on $\pi(H_3^r)$ for small r. Independently of [50], Gunderson and Semeraro [29, Theorem 1.1] produced different constructions for $r \in \{6, 7, 8\}$.

Let us turn our attention to the case when $r \to \infty$. Frankl and Füredi [21, Example 3] observed that $\pi(H_3^r)$ is at least $(1 + o(1))/2^{r-1}$ as $r \to \infty$. Sidorenko [50, Corollary 4.6] proved that $\pi(H_3^r) \ge (1.7215 - o(1))/r^2$ and Clemen [12, Theorem 1.2] improved this to $\pi(H_3^r) = \Omega(r^{-2}\sqrt{\log r})$.

The first main result of this paper determines the order of magnitude of $\pi(H_3^r)$. Since the construction easily adapts to work for H_k^r with any constant k, we present this more general result.

Theorem 1.1. *For a positive integer h, define*

$$c_h := \sup_{\lambda > 0} \lambda \mathrm{e}^{-\lambda} \sum_{j=0}^{h-1} \frac{\lambda^j}{j!}.$$

Then, for every fixed $k \ge 3$, we have

$$\liminf_{r \to \infty} r\pi(H_k^r) \ge c_{\lfloor (k-1)/2 \rfloor}. \tag{1.2}$$

In the case $k = 3$ of Theorem 1.1, we have $c_1 = \mathrm{e}^{-1}$ (with the supremum attained at $\lambda := 1$), so we obtain that $\pi(H_3^r) \ge (\mathrm{e}^{-1} - o(1))/r$. Thus we know $\pi(H_3^r)$ within a factor $\mathrm{e} + o(1)$ as $r \to \infty$.

For general H_k^r, Sidorenko [50] gave a geometric construction proving that

$$\pi(H_k^r) \ge \mathbb{E}(\xi_{r,k-1}), \quad \text{for all } 3 \le k \le r+1, \tag{1.3}$$

that is, $\pi(H_k^r)$ is at least the expectation of $\xi_{r,k-1}$, where for $2 \le m \le r$ the random variable $\xi_{r,m}$ is the shortest length of an arc that contains at least m points among r independent uniformly distributed points on the circle of circumference 1. For each fixed $k \ge 3$, it can be derived from the results of Cressie [13] that $\mathbb{E}(\xi_{r,k-1}) = (C_k + o(1))r^{-1-\frac{1}{k-2}}$ with an explicit constant C_k; by (1.3) this is a lower bound on $\pi(H_k^r)$. The value of $\mathbb{E}(\xi_{r,k-1})$ for growing $k = k(r)$ was not estimated in [50] in the full range since direct computations become tedious.

By using the exponential representation for consecutive arc lengths and the maximal Bernstein inequality (instead of working with r-fold integrals), we prove a lower bound on $\mathbb{E}(\xi_{r,k-1})$ which translates via (1.3) into the following lower bound on $\pi(H_k^r)$.

Theorem 1.2. *For all integers $r \ge 2$ and $3 \le k \le r+1$, it holds that*

$$\pi(H_k^r) \ge \frac{k-2}{r} - \frac{24}{r}\left(\sqrt{s \log \frac{\mathrm{e}r}{s}} + \log \frac{\mathrm{e}r}{s}\right),$$

where $s := \min\{k-2, r-k+2\}$.

To illustrate the obtained lower bound, we write its direct consequences in some broad ranges of $k = k(r)$, also combining them with the upper bound from (1.1).

Corollary 1.3. *Let $k = k(r)$ be an integer with $3 \le k \le r+1$.*

(i) *If $k/\log r \to \infty$, then*

$$\pi(H_k^r) = (1+o(1))\frac{k}{r}.$$

(ii) *If $k/r \to \alpha \in (0,1)$ and $\beta := \min\{\alpha, 1-\alpha\}$, then*

$$0 \le \frac{k-2}{r} - \pi(H_k^r) \le \frac{24\sqrt{\beta\log(\mathrm{e}/\beta)} + o(1)}{\sqrt{r}}.$$

Thus, if k grows faster than $\log r$ then we know the value of $\pi(H_k^r)$ asymptotically. Also, the error term can be made more precise for larger k, with an example of such an estimate given by Item (ii) above.

In the remaining range $k = O(\log r)$, one can get bounds that are within a constant factor of each other by taking the unions of r-graphs returned by Theorems 1.1 and 1.2; specifically, the bounds are as follows.

Corollary 1.4. *Let $k = k(r)$ be an integer with $3 \le k \le r+1$. Then for every $\varepsilon > 0$ there is r_0 such that, for all $r \ge r_0$,*

$$\pi(H_k^r) \ge \left(\frac{1}{2\mathrm{e}} - \varepsilon\right)\frac{k-2}{r}. \tag{1.4}$$

Moreover, if $k(r) \to \infty$ as $r \to \infty$, then, for all sufficiently large r, (1.4) can be strengthened to $\pi(H_k^r) \ge (\frac{1}{2} - \varepsilon)\frac{k}{r}$.

Thus, by (1.1), we know $\pi(H_k^r)$ within a multiplicative factor $2\mathrm{e}+o(1)$ uniformly in $k \in \{3, \dots, r+1\}$ and this factor can be reduced to $2+o(1)$ if $k \to \infty$. Independently of us, a proof of $\pi(H_k^r) \ge ck/r$ for a constant $c > 0$ was obtained by Pickhardt [41]. (Also, some weaker versions of Theorems 1.1 and 1.2 appeared previously in a preprint [55] by the last two authors; this preprint is now fully superseded by the current paper.)

Next, we turn our attention to another important class of sequences of forbidden r-graphs. For a fixed t-graph F and an integer $r \ge t$, let the *r-augmentation* $F(r)$ of F be the r-graph obtained by adjoining the same set of $r-t$ new vertices to every edge of F. This construction is also known in the literature under other names (*r-suspension*, *$(r-t)$-fold enlargement*, *(iterated) cone*, etc). There is a large number of results on the Turán function of these r-graphs and we do not attempt to survey them here. A connection to the previous results comes from the observation that H_k^r is the r-augmentation of K_k^{k-1}. The augmentations of general complete t-graphs K_k^t are called *daisies*, a term introduced by Bollobás, Leader and Malvenuto [6] who initiated a systematic study of the Turán densities of daisies. A breakthrough in this area was recently obtained by Ellis, Ivan and Leader [16, Theorem 4] who proved, in particular, that $\pi(K_4^2(r))$ is bounded away from 0. This result and the underlying method have already found many applications, including those in the original paper [16] and the subsequent applications found by Alon [1] (to erasure list-decodable codes), Grebennikov and Marciano [27] (to edge-sets in hypercubes avoiding a 10-cycle), van der Pol, Walsh and Wigal [54] (to maximising the number of bases in matroids), Axenovich and Liu [2] (to visibility in hypercubes), etc.

If $t = 2$, that is, F is a graph, then the proof in [16] shows that $\pi(F(r))$ is bounded away from 0 not only when $F = K_4^2$ is the 4-clique but also when F is an arbitrary graph of chromatic number at least 4 (see [16, Page 3852]). If F is a bipartite graph then the r-graph $F(r)$ is r-partite, which implies by the classical result of Erdős [17] that $\pi(F(r)) = 0$. If F is 3-chromatic then e.g. the more general result of Cheng, Gerbner, Hama Karim and Zhou [10, Theorem 1.2] implies that $\pi(F(r)) \le \pi(H_3^r)$. Thus, by (1.1), we have $\pi(F(r)) \le 1/r$.

The following result shows that this is tight up to a factor $(\log r)^{1+o(1)}$.

Theorem 1.5. *For every fixed graph F of chromatic number 3, it holds that*

$$\pi(F(r)) \ge \frac{1}{r(\log r)^{1+o(1)}}, \quad \text{as } r \to \infty.$$

When one analyses the proof of Theorem 1.5 and tries to abstract the properties of the graph F that are needed, the following property of general t-graphs naturally appears. We say that a t-graph F has *zero kernel* if its edge–vertex incidence matrix B_F has zero kernel (or, equivalently, full column rank) over $\mathbb{R}$. It is easy to see (see Lemma 5.8) that every connected non-bipartite graph F has zero kernel. Since, every graph of chromatic number at least 3 has such a component, Theorem 1.5 follows from the following more general result.

Theorem 1.6. *Let $t \geq 2$ and F be a fixed non-empty t-graph such that B_F has zero kernel over $\mathbb{R}$. Then*

$$\pi(F(r)) \geq \frac{1}{r(\log r)^{1+o(1)}}, \quad \text{as } r \to \infty.$$

This is sharp up to a $(\log r)^{1+o(1)}$ factor if, for example, the t-graph F admits a homomorphism to K^t_{t+1} (that is, admits a vertex colouring with $t+1$ colours in which every edge is rainbow). Indeed, then the r-augmentation $F(r)$ admits a homomorphism to $K^t_{t+1}(r) = H^r_{t+1}$. This implies via super-saturation (see e.g. [34, Theorem 2.2]) that $\pi(F(r)) \leq \pi(H^r_{t+1})$, which by (1.1) is at most $(t-1)/r = O(1/r)$.

Organization of the paper. Section 2 defines some commonly used terms and notions. In Section 3, we prove Theorem 1.1 using a probabilistic construction involving finite geometries. In Section 4, we describe Sidorenko's circular construction and then prove Theorem 1.2 and Corollary 1.3 (that estimate the lower bounds on $\pi(H^r_k)$ coming from the construction) as well as Corollary 1.4 (that extends the bounds to the remaining range $k = O(\log r)$). Theorem 1.6 is proved in Section 5 via a different probabilistic construction. As we already mentioned, Theorem 1.5 directly follows from Theorem 1.6 when combined with the easy fact (proved in Lemma 5.8) that every connected non-bipartite graph has the zero kernel property. Finally, Section 6 contains some concluding remarks and open problems.

2 Some notation and conventions

For a positive integer n, we denote $[n] := \{1, \dots, n\}$ and let $\mathbb{Z}_n$ denote the additive group of residues modulo n. For a prime power q, let $\mathbb{F}_q$ denote the finite field with q elements. For integers $n, r \geq 0$, the *falling factorial* is $(n)_r := \prod_{i=0}^{r-1}(n-i)$. An empty product is assumed to have value 1. When we write log, we mean the natural logarithm (i.e. with base e = 2.718...).

The *argument* (or *angle*) of a non-zero complex number $z = \rho e^{i\theta} \in \mathbb{C}$ with $(\rho, \theta) \in (0, \infty) \times [0, 2\pi)$ is $\arg(z) := \theta$; thus its value lies in $[0, 2\pi)$.

The *binomial distribution* $\mathrm{Bin}(n, p)$ takes value $i \in \{0, \dots, n\}$ with probability $\binom{n}{i} p^i (1-p)^{n-i}$. The probability and the expectation of a random variable are denoted by $\mathbb{P}$ and $\mathbb{E}$ respectively.

Let G be an r-graph. Its vertex and edge sets are denoted by $V(G)$ and $E(G)$. The number of edges in G is denoted by $|G|$. We call G *non-empty* if $|G| > 0$. Denoting the vertices of G by $v_1, \dots, v_n$, a *blowup* of G is an r-graph G' on the union of some pairwise disjoint (possibly empty) sets $V_1, \dots, V_n$ where, for every edge $\{v_{i_1}, \dots, v_{i_r}\} \in G$, we add the complete r-partite r-graph with parts $V_{i_1}, \dots, V_{i_r}$ to the edge set of G'.

Let $\mathbf{1}$ (resp. $\mathbf{0}$) denote the all-1 (resp. all-0) vector; its index set should always be clear from the context. The *kernel* $\{\boldsymbol{v} : M\boldsymbol{v} = \mathbf{0}\}$ of a matrix M is denoted by $\ker M$. Unless stated otherwise, the underlying field in linear algebra statements is $\mathbb{R}$.

3 Lower bounds on H^r_k via caps and random maps

In this section, we prove Theorem 1.1, using random linear maps and known constructions of cap sets over the finite field $\mathbb{F}_q$ for some prime power $q = q(r, k)$.

We will prove the following result (that has all parameters explicitly written) and then derive Theorem 1.1 from it.

Theorem 3.1. *Let q be a prime power and let $N \geq r \geq 2$ and $3 \leq k \leq r+1$ be integers. Define*

$h := \lfloor (k-1)/2 \rfloor$ *and*

$$\begin{aligned} p_{r,h}(q) &:= \left(\prod_{i=2}^{r-1}(1-q^{-i})\right)\mathbb{P}\big(\mathrm{Bin}(r,1/q) \le h-1\big), \\ s(q) &:= \frac{(q-1)(q^2+1)}{q^4}. \end{aligned}$$

Then there is an H_k^r-free r-graph G on N vertices with at least

$$\binom{N}{r} p_{r,h}(q)s(q)(1-q^{-3})^{N-r}$$

edges. Consequently,

$$\pi(H_k^r) \ge \frac{(N)_r}{N^r}\, p_{r,h}(q)s(q)(1-q^{-3})^{N-r}. \tag{3.1}$$

3.1 Construction via caps and random linear maps

Fix q, N, r, k as in Theorem 3.1. Recall that $h = \lfloor (k-1)/2 \rfloor$. We will use the projective geometry $\mathrm{PG}(3,q)$, whose points are the 1-dimensional subspaces of $\mathbb{F}_q^4$ (called *projective points*). A *projective line* in $\mathrm{PG}(3,q)$ consists of the projective points in a 2-dimensional subspace of $\mathbb{F}_q^4$. We write $[\boldsymbol{v}]$ for the projective point represented by a non-zero vector $\boldsymbol{v} \in \mathbb{F}_q^4$.

A *cap* is a set of points with no three collinear. We use the following classical result of Bose [7, Eq. (5.60)]. (For modern texts containing a proof, see e.g. [3, Theorem 4.36] or [9, Proposition 2.6.6].)

Lemma 3.2 (Bose [7])**.** *For every prime power q, there is a cap $\mathcal{C}_q \subseteq \mathrm{PG}(3,q)$ with q^2+1 points.*

Fix such a cap $\mathcal{C}_q \subseteq \mathrm{PG}(3,q)$ and let

$$S_q := \{\boldsymbol{v} \in \mathbb{F}_q^4 \setminus \{\mathbf{0}\} : [\boldsymbol{v}] \in \mathcal{C}_q\}$$

be its *cone*. Each projective point has $q-1$ non-zero representatives, so $|S_q| = (q-1)(q^2+1)$ and $|S_q|/q^4 = s(q)$.

Choose independent uniform random linear maps

$$M : \mathbb{F}_q^N \to \mathbb{F}_q^{r-1} \quad \text{and} \quad L : \mathbb{F}_q^N \to \mathbb{F}_q^4,$$

equivalently, the entries of their matrices are all independent and uniform in $\mathbb{F}_q$. We use the following standard fact.

Lemma 3.3. *Let V, W be finite-dimensional vector spaces over $\mathbb{F}_q$, and choose a uniform linear map $T : V \to W$. If $\boldsymbol{v}_1, \dots, \boldsymbol{v}_s$ are linearly independent in V, then $T\boldsymbol{v}_1, \dots, T\boldsymbol{v}_s$ are independent uniform elements of W.*

Proof. Extend the given vectors to a basis of V. A linear map is uniquely determined by its images on this basis, and all choices of these images occur with equal probability. □

For $A \subseteq [N]$, let $\mathbb{F}_q^A \subseteq \mathbb{F}_q^N$ be the coordinate subspace supported on A, and let the *kernel* of A be

$$P_A := \ker M \cap \mathbb{F}_q^A,$$

that is, the subspace $P_A \subseteq \mathbb{F}_q^N$ consists of the vectors that are sent to $\mathbf{0}$ by M and whose coordinates in $[N] \setminus A$ are all 0.

For an r-set A, we say that A is *admissible* if $\dim P_A = 1$ and a non-zero generator $\boldsymbol{c}_A$ of P_A has at most $h-1$ zero coordinates indexed by A. This condition does not depend on the choice of the generator since we require that P_A is 1-dimensional. Choose $\boldsymbol{c}_A$ as a function of M alone. When $h = 1$, an admissible set is a *circuit*: its kernel P_A is generated by a vector supported on all of A.

If A is admissible and $A \subseteq B \subseteq [N]$, then the $(r-1) \times r$ submatrix $M|_{\mathbb{F}_q^A}$ of M, where we take the columns indexed by A, has the maximum possible rank $r-1$, so $M|_{\mathbb{F}_q^B}$ also has rank $r-1$. Thus

$$\dim P_B = |B| - r + 1. \tag{3.2}$$

Define the initial r-graph G_0 by

$$A \in G_0 \quad \iff \quad A \text{ is admissible and } L\boldsymbol{c}_A \in S_q.$$

Call an $(r+1)$-set U *bad* if $L(P_U)$ is a 1-dimensional subspace representing a point of $\mathcal{C}_q$. Delete from G_0 every edge contained in any bad set, and denote the resulting r-graph by G.

Before proving that the obtained graph G is always H_k^r-free, let us informally discuss this construction. Its most interesting ingredient is probably the map that sends some r-sets A (in fact, a positive fraction of them in our final choice of parameters) to vectors $\boldsymbol{c}_A$ such that, for every $(r+1)$-set U, all resulting vectors are confined to a subspace (namely, P_U) whose dimension is 2 unless some degeneracy occurs in $M|_{\mathbb{F}_q^U}$ (which is rare so we can just remove all involved r-sets).[1] Thus if we take the image of these vectors under the random map L then those coming from any given $(r+1)$-set U can hit at most two different elements of the cap $\mathcal{C}_q$ and we just need to control the multiplicity of each preimage.

Let us give a formal proof of the H_k^r-freeness of G, which is actually quite short.

Lemma 3.4. *For every outcome of the two random maps M and L, the resulting r-graph G is H_k^r-free.*

Proof. Fix an $(r+1)$-set U containing at least one edge of G_0. By (3.2), we have $\dim P_U = 2$.

First, we show that any fixed 1-dimensional subspace of P_U can be the kernel of at most h admissible r-subsets of U. Suppose that this subspace is P_A, where $A = U \setminus \{u\}$ is admissible. Its generator $\boldsymbol{c}_A$ has zero coordinate at u and at most $h-1$ zero coordinates indexed by A, so it has at most h zero coordinates indexed by U. If some r-set $A' = U \setminus \{u'\}$ has $P_{A'} = P_A$, then the u'-coordinate of $\boldsymbol{c}_A$ is zero. Hence there are at most h possibilities for A', as claimed.

If $\dim L(P_U) = 2$, then $L|_{P_U}$ is injective. For distinct subspaces P_A, where A ranges over all initial edges in U, they are mapped to distinct projective points of $\mathcal{C}_q$. Since $\mathcal{C}_q$ is a cap, there are at most two such subspaces P_A. Each accounts for at most h initial edges, so U contains at most $2h$ edges of G_0.

The case $\dim L(P_U) = 0$ is impossible, since an initial edge $A \subseteq U$ forces a non-zero vector $L\boldsymbol{c}_A \in L(P_U)$. If $\dim L(P_U) = 1$, then the presence of an initial edge implies that the projective point corresponding to $L(P_U)$ belongs to $\mathcal{C}_q$. Thus U is bad and all initial edges inside U are deleted.

In every case, U contains at most $2h \le k-1$ edges of G, as required. □

3.2 Determining the expected size of G

Next, for a fixed r-set $A \subseteq [N]$, we write an explicit formula for the probability that A is an edge of G.

We first compute the probability that A is admissible. The $(r-1) \times r$ submatrix $M|_{\mathbb{F}_q^A}$ of M has rank $r-1$ with probability

$$\prod_{j=0}^{r-2} \left(1 - \frac{q^j}{q^r}\right) = \prod_{i=2}^{r} (1 - q^{-i}),$$

as follows by exposing its rows successively and requiring linear independence. Conditional on this rank, its kernel is a uniform 1-dimensional subspace of $\mathbb{F}_q^A$. Indeed, right multiplication by any invertible $r \times r$ matrix preserves the distribution of the submatrix and acts transitively on its possible kernel lines.

A uniform kernel line has the same distribution of zero coordinates as a uniform non-zero vector in $\mathbb{F}_q^A$, since every line has $q-1$ non-zero vectors. The number of zero coordinates of a uniform vector in $\mathbb{F}_q^A$ (without the non-zero conditioning) is distributed as $Z \sim \mathrm{Bin}(r, 1/q)$. Since $h - 1 < r$, the event $Z \le h-1$ excludes the zero vector. Hence

$$\mathbb{P}(A \text{ is admissible}) = \left(\prod_{i=2}^{r} (1 - q^{-i})\right) \frac{\mathbb{P}(Z \le h-1)}{1 - q^{-r}} = p_{r,h}(q).$$

[1] Our literature search returned that the idea of placing circuit vectors supported on a set U in a subspace of dimension $|U| - \mathrm{rank}(M|_{\mathbb{F}_q^U})$ was used by e.g. Freij-Hollanti, Jurrius and Kuznetsova [23, Eqs. (1)–(2)].

Next, condition on the entire matrix M such that A is admissible. By Lemma 3.3, the image $L\boldsymbol{c}_A$ is uniform in $\mathbb{F}_q^4$, so it belongs to S_q with probability $s(q)$. For each $x \in [N] \setminus A$, choose a vector $\boldsymbol{u}^x \in P_{A\cup\{x\}}$ with x-coordinate equal to 1, making this choice as a function of M alone. Such a vector exists: the coordinate projection from $P_{A\cup\{x\}}$ (which has dimension 2 by (3.2)) to $\mathbb{F}_q^{\{x\}}$ has 1-dimensional kernel P_A so its image is 1-dimensional and contains the whole of $\mathbb{F}_q^{\{x\}}$.

The vectors

$$\{\boldsymbol{c}_A\} \cup \{\boldsymbol{u}^x : x \in [N] \setminus A\}$$

are linearly independent: in any linear relation between them, the coordinate at $x \in [N] \setminus A$ is the coefficient of $\boldsymbol{u}^x$, and once all these coefficients vanish, the coefficient of $\boldsymbol{c}_A$ vanishes too. Their images under L are therefore independent uniform vectors by Lemma 3.3.

Condition further on $L\boldsymbol{c}_A = \boldsymbol{y}$ for a fixed $\boldsymbol{y} \in S_q$. The vectors $L\boldsymbol{u}^x$ remain independent and uniform in $\mathbb{F}_q^4$. Since

$$P_{A\cup\{x\}} = \operatorname{span}\{\boldsymbol{c}_A, \boldsymbol{u}^x\},$$

the set $A\cup\{x\}$ is bad exactly when $L\boldsymbol{u}^x \in \operatorname{span}\{\boldsymbol{y}\}$. This has probability $q/q^4 = q^{-3}$, independently for each $x \in [N] \setminus A$. These are all the $(r+1)$-sets containing A. Consequently,

$$\mathbb{P}(A \in G) = \mathbb{P}(A \text{ is admissible}) \, \frac{|S_q|}{q^4} \, (1-q^{-3})^{N-r} = p_{r,h}(q)s(q)(1-q^{-3})^{N-r}.$$

Summing over all r-sets A yields

$$\mathbb{E}\,|G| = \binom{N}{r} p_{r,h}(q)s(q)(1-q^{-3})^{N-r}. \tag{3.3}$$

3.3 Deriving Theorems 3.1 and 1.1

Proof of Theorem 3.1. By Lemma 3.4, every outcome of G is H_k^r-free. Choose an outcome whose edge count is at least the expectation in (3.3). This proves the assertion about the number of edges.

Every pair of vertices of H_k^r lies in an edge (since $k \geq 3$). Consequently, any copy of H_k^r in a blowup of G would use every part at most once and would project to a copy of H_k^r in G. Thus all blowups of G are also H_k^r-free. Replacing each vertex of G by m clones and letting $m \to \infty$ gives

$$\pi(H_k^r) \geq \lim_{m\to\infty} \frac{|G|m^r}{\binom{Nm}{r}} = \frac{r!|G|}{N^r} \geq \frac{(N)_r}{N^r} \, p_{r,h}(q)s(q)(1-q^{-3})^{N-r},$$

which proves (3.1). □

Next, we derive Theorem 1.1 from Theorem 3.1. We will use the following elementary inequality:

$$\prod_{i=1}^{m}(1-a_i) \geq 1 - \sum_{i=1}^{m} a_i, \quad \text{for any } a_1, \ldots, a_m \in [0,1], \tag{3.4}$$

which follows by induction on m.

Proof of Theorem 1.1. Fix $k \geq 3$, put $h := \lfloor (k-1)/2 \rfloor$, and fix $\lambda > 0$. Let $r \to \infty$. Let q be the least prime with $q \geq r/\lambda$. The Prime Number Theorem implies that $q = (1+o(1))r/\lambda$. Put

$$N := \lceil r^{5/2} \rceil.$$

For the binomial random variable $Z \sim \operatorname{Bin}(r, 1/q)$ and each fixed non-negative integer j, we have

$$\mathbb{P}(Z = j) = \binom{r}{j} q^{-j}(1-q^{-1})^{r-j} \to \mathrm{e}^{-\lambda}\frac{\lambda^j}{j!}.$$

Also, by (3.4),

$$1 \geq \prod_{i=2}^{r-1}(1-q^{-i}) \geq 1 - \sum_{i=2}^{\infty} q^{-i} = 1 - \frac{1}{q(q-1)} \to 1,$$

while

$$rs(q) = \frac{r}{q}(1-q^{-1})(1+q^{-2}) \to \lambda.$$

Two further applications of (3.4) give

$$1 \geq \frac{(N)_r}{N^r} \geq 1 - \frac{r(r-1)}{2N} \to 1,$$
$$1 \geq (1-q^{-3})^{N-r} \geq 1 - (N-r)q^{-3} \to 1.$$

Substituting these limits into (3.1) yields

$$\liminf_{r\to\infty} r\pi(H_k^r) \geq \lambda \mathrm{e}^{-\lambda} \sum_{j=0}^{h-1} \frac{\lambda^j}{j!}.$$

Since $\lambda > 0$ was arbitrary, taking the supremum proves (1.2). □

4 Lower bounds from the circular construction

In this section, we prove Theorem 1.2, where we estimate the lower bound on $\pi(H_k^r)$ coming from the circular construction of Sidorenko [50].

We first recall this construction. Let

$$\mathcal{C} := \{z \in \mathbb{C} : |z| = 1\}$$

be the unit circle in the complex plane. Let $G_{r,k-1}$ be the (infinite) r-graph with vertex set $\mathcal{C}$, where an r-subset $A \subseteq \mathcal{C}$ is an edge if and only if

$$\arg\Big(\prod_{z\in A} z\Big) \in [0, \Delta_{k-1}(A)]\,,$$

where $\Delta_m(A)$ denotes the shortest length of a closed arc in $\mathcal{C}$ that contains at least m elements of A. Thus an r-set A is an edge of $G_{r,k-1}$ if the sum of the angles of its elements modulo 2π is at most $\Delta_{k-1}(A)$. Sidorenko [50, Theorem 3.1] proved that this r-graph is H_k^r-free.

Let $z_1, \dots, z_r$ be r independent random points uniformly distributed on the circle $\mathcal{C}$ and let $A := \{z_1, \dots, z_r\}$; these points are almost surely distinct. When we apply a uniformly random rotation ϕ of the circle $\mathcal{C}$ to any fixed outcome of distinct points $z_1, \dots, z_r$, the rotated r-set $\phi(A)$ is an edge of $G_{r,k-1}$ with probability

$$\xi_{r,k-1} = \xi_{r,k-1}(z_1, \dots, z_r) := \frac{\Delta_{k-1}(A)}{2\pi}.$$

This holds since $\arg(\prod_{j=1}^r \phi(z_j))$ is uniformly distributed in $[0, 2\pi)$ while $\Delta_{k-1}(\phi(A)) = \Delta_{k-1}(A)$ for any rotation ϕ.

By considering the subgraph of $G_{r,k-1}$ induced by $n \to \infty$ independent uniformly distributed points on the circle and using the Fubini–Tonelli theorem, we obtain Inequality (1.3) from the Introduction (which states that $\pi(H_k^r)$ is at least the expectation of the random variable $\xi_{r,k-1}$). Thus our task is to estimate $\mathbb{E}(\xi_{r,k-1})$ for independent uniform $z_1, \dots, z_r \in \mathcal{C}$.

Let $z_1' = z_1$, and relabel the remaining points as $z_2', \dots, z_r'$ in counterclockwise order starting from z_1. When working with the variables z_i' and various cyclic sums, we identify the index set $[r] = \{1, \dots, r\}$ with $\mathbb{Z}_r$ and do index addition modulo r. Let D_i be the length of the counterclockwise arc from z_i' to z_{i+1}' for $i \in [r]$. Then

$$\Delta_{k-1}(A) = \min_{1\leq i\leq r} \sum_{j=0}^{k-3} D_{i+j}.$$

Clearly, $D_1 + \cdots + D_r = 2\pi$.

The distribution of $(D_1, \dots, D_r)$ can be described using the following standard *normalized exponential representation* (see Pyke [44, Section 4.1] and compare with the related order-statistic

representations of Rényi [45]). Let $X_1, \dots, X_r$ be independent exponential random variables with mean 1, that is, each has density e^{-x} for $x \geq 0$; extend their indices cyclically modulo r. Let $S_r := \sum_{j=1}^{r} X_j$. For $1 \leq \ell \leq r-1$ and $i \in [r]$, we define

$$Y_i^{(\ell)} := \sum_{j=0}^{\ell-1} X_{i+j} \quad \text{and} \quad M_{r,\ell} := \min_{1 \leq i \leq r} Y_i^{(\ell)}.$$

The following result is standard; the spacing density used in its proof appears, for example, in Hajós and Rényi [30, Eq. (6)]. For completeness, we give a proof.

Lemma 4.1. *The vector* $\left(\frac{D_1}{2\pi}, \dots, \frac{D_r}{2\pi}\right)$ *has the same distribution as* $\left(\frac{X_1}{S_r}, \dots, \frac{X_r}{S_r}\right)$*, and the latter normalized vector is independent of* S_r*. Consequently, for* $1 \leq m \leq r-1$*, we have*

$$\mathbb{E}(\xi_{r,m+1}) = \frac{1}{r}\,\mathbb{E}(M_{r,m}). \tag{4.1}$$

Proof. Conditional on z_1, the counterclockwise angles of the other $r-1$ points relative to z_1, divided by 2π, are independent uniform variables on $[0,1)$. Their order statistics have joint density $(r-1)!$ on $0 < u_1 < \cdots < u_{r-1} < 1$. The change of variables from these order statistics to the first $r-1$ consecutive spacings has Jacobian determinant 1. Thus $(D_1/(2\pi), \dots, D_r/(2\pi))$ has constant density $(r-1)!$ on the simplex of positive coordinates summing to 1, with respect to its first $r-1$ coordinates.

For $j \in [r-1]$, write $v_j := X_j/s$, where $s := \sum_{i=1}^{r} X_i$, and let $v_r := 1 - \sum_{i=1}^{r-1} v_i$. The inverse change of variables, $X_i = s v_i$ for $i \in [r]$, has Jacobian determinant s^{r-1}, so the joint density of $(v_1, \dots, v_{r-1}, s)$ is

$$s^{r-1} \prod_{i=1}^{r} \mathrm{e}^{-s v_i} = \frac{s^{r-1}\mathrm{e}^{-s}}{(r-1)!}\,(r-1)!.$$

The support is the product of the simplex described above and $(0, \infty)$. This factorization proves both the asserted distribution and the independence from S_r.

Finally, for $\boldsymbol{v} = (v_1, \dots, v_r)$, define $f(\boldsymbol{v}) := \min_{i \in [r]} \sum_{j=0}^{m-1} v_{i+j}$. Then $M_{r,m} = S_r f((X_1/S_r, \dots, X_r/S_r))$ and $f((X_1/S_r, \dots, X_r/S_r))$ has the distribution of $\xi_{r,m+1}$. Independence and $\mathbb{E}S_r = r$ give the claimed expectation identity (4.1). □

Let $Z_j := X_j - 1$ for $j \in [r]$, so the variables Z_j are independent, with mean 0 and variance 1. We will apply the maximal Bernstein inequality to the centred exponential variables Z_j. We will use the following version that can be obtained, for example, from [46, Theorem B.2] (using $K = 1$ there and applying it separately to the variables Z_j and their negations $-Z_j$).

Lemma 4.2. *Let* $n \geq 1$ *be an integer, let* $Z_1, \dots, Z_n$ *be independent copies of* $X_1 - 1$*, and let* $T_j := \sum_{i=1}^{j} Z_i$*. Then, for every* $u \geq 0$*, we have*

$$\mathbb{P}\left(\max_{1 \leq j \leq n} |T_j| \geq u\right) \leq 2\exp\left[-\frac{u^2}{2(n+u)}\right] \leq 2\exp\left[-\frac{1}{4}\min\left\{\frac{u^2}{n}, u\right\}\right].$$

The main part of the proof lies in applying the above lemma to estimate the maximal deviation of a sum of consecutive variables Z_j from its mean.

Lemma 4.3. *Let* ℓ *be an integer with* $1 \leq \ell \leq r/2$*. Define* $L_\ell := \log(\mathrm{e}r/\ell)$ *and, for* $i \in [r]$*,* $W_i^{(\ell)} := \sum_{j=0}^{\ell-1} Z_{i+j}$*. Then*

$$\mathbb{E}\left(\max_{1 \leq i \leq r} |W_i^{(\ell)}|\right) \leq 24\left(\sqrt{\ell L_\ell} + L_\ell\right).$$

Proof. To estimate the maximum absolute value of the sums $W_i^{(\ell)}$ for $1 \leq i \leq r$, we group them according to their starting index i. Namely, fix a partition of $[r]$ into $\lceil r/\ell \rceil$ consecutive blocks, each containing at most ℓ indices. The number of blocks $\lceil r/\ell \rceil$ is at most $2r/\ell$, since $\ell \leq r/2$. Take a block $B = \{a, a+1, \dots, a+d-1\}$, where $d \leq \ell$. The cyclic sum beginning with Z_i for some $i \in B$ can involve only the variables $Z_a, \dots, Z_{a+d+\ell-2}$. Their number is $d + \ell - 1 \leq 2\ell - 1 < r$. Thus, no

variable occurs twice in this sequence. For convenience, relabel these distinct variables as $Z'_1, \dots, Z'_n$ (that is, $Z'_i = Z_{i+a-1}$ for $i \in [n]$), where

$$n := d + \ell - 1 \le 2\ell,$$

and define $Q_0 := 0$ and $Q_j := \sum_{h=1}^{j} Z'_h$ for $1 \le j \le n$ (see Figure 1).

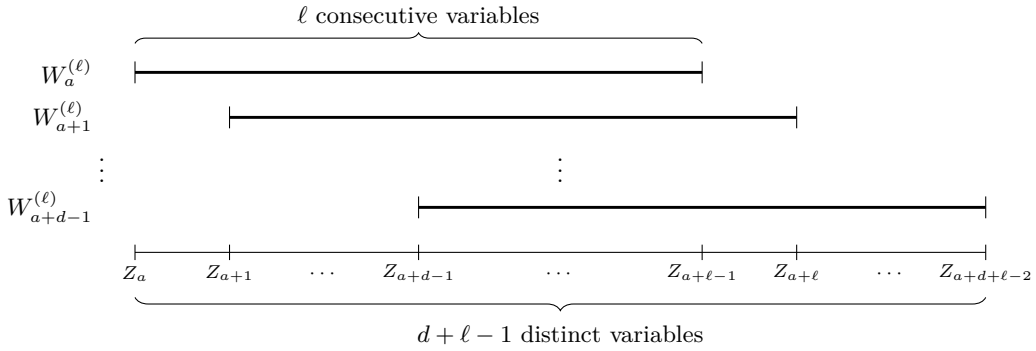


Figure 1: One block of starting indices: $a, a+1, \dots, a+d-1$

For each $0 \le s \le d-1$, the cyclic sum beginning at $a+s \in B$ is

$$W_{a+s}^{(\ell)} = Z'_{s+1} + \cdots + Z'_{s+\ell} = Q_{s+\ell} - Q_s.$$

Since $s + \ell \le d - 1 + \ell = n$, both $Q_{s+\ell}$ and Q_s are among $Q_0, \dots, Q_n$ and $|W_{a+s}^{(\ell)}| \le |Q_{s+\ell}| + |Q_s|$. Hence

$$\max_{i \in B} |W_i^{(\ell)}| \le 2 \max_{0 \le j \le n} |Q_j|.$$

It follows that, if the left-hand side is at least t, then $\max_{0 \le j \le n} |Q_j| \ge t/2$. For $t \ge 0$, applying Lemma 4.2 with $u = t/2$ gives

$$\begin{aligned} \mathbb{P}\left(\max_{i \in B} |W_i^{(\ell)}| \ge t \right) &\le \mathbb{P}\left(\max_{0 \le j \le n} |Q_j| \ge t/2 \right) \\ &\le 2 \exp\left(-\frac{1}{4} \min\left\{ \frac{t^2}{4n}, \frac{t}{2} \right\} \right) \le 2 \exp\left(-\min\left\{ \frac{t^2}{32\ell}, \frac{t}{8} \right\} \right), \end{aligned}$$

where the last inequality uses that $n \le 2\ell$.

Let $Y := \max_{1 \le i \le r} |W_i^{(\ell)}|$. There are at most $2r/\ell$ blocks, so the union bound gives

$$\begin{aligned} \mathbb{P}(Y \ge t) &\le \min\left\{ 1, \sum_{\text{block } B} \mathbb{P}\left(\max_{i \in B} |W_i^{(\ell)}| \ge t \right) \right\} \\ &\le \min\left\{ 1, \frac{4r}{\ell} \exp\left(-\min\left\{ \frac{t^2}{32\ell}, \frac{t}{8} \right\} \right) \right\}. \end{aligned}$$

We now estimate the expectation $\mathbb{E}(Y)$. Since

$$\exp\left(-\min\left\{ \frac{t^2}{32\ell}, \frac{t}{8} \right\} \right) \le \exp\left(-\frac{t^2}{32\ell} \right) + \exp\left(-\frac{t}{8} \right)$$

and $\min\{1, x+y\} \le \min\{1, 2x\} + \min\{1, 2y\}$ for all $x, y \ge 0$, we obtain

$$\mathbb{E}(Y) = \int_0^\infty \mathbb{P}(Y \ge t)\, dt$$

$$\leq \int_0^\infty \min\left\{1, \frac{8r}{\ell}\exp\left(-\frac{t^2}{32\ell}\right)\right\} dt + \int_0^\infty \min\left\{1, \frac{8r}{\ell}\exp\left(-\frac{t}{8}\right)\right\} dt.$$

Let $t_0 := \sqrt{32\ell\log(8r/\ell)}$. Then $(8r/\ell)\exp(-t_0^2/(32\ell)) = 1$. Splitting the first integral at t_0 and using $\int_x^\infty \mathrm{e}^{-av^2}\,dv \leq \mathrm{e}^{-ax^2}/(2ax)$ for $a, x > 0$, we obtain

$$\int_0^\infty \min\left\{1, \frac{8r}{\ell}\exp\left(-\frac{t^2}{32\ell}\right)\right\} dt \leq \sqrt{32\ell\log\frac{8r}{\ell}} + \frac{16\ell}{\sqrt{32\ell\log(8r/\ell)}}.$$

Similarly, let $t_1 := 8\log(8r/\ell)$ and split the second integral at t_1 to get

$$\int_0^\infty \min\left\{1, \frac{8r}{\ell}\exp\left(-\frac{t}{8}\right)\right\} dt \leq 8\left(\log\frac{8r}{\ell} + 1\right).$$

Recall that $L_\ell = \log(\mathrm{e}r/\ell)$. Since $\ell \leq r/2$, we have $8r/\ell \geq 16$, $L_\ell \geq \log(2\mathrm{e}) > 1$, and $\log(8r/\ell) \leq 2L_\ell$. It follows that

$$\sqrt{32\ell\log\frac{8r}{\ell}} \leq 8\sqrt{\ell L_\ell} \quad \text{and} \quad \frac{16\ell}{\sqrt{32\ell\log(8r/\ell)}} \leq 2\sqrt{\ell} \leq 2\sqrt{\ell L_\ell}.$$

Moreover,

$$8\left(\log\frac{8r}{\ell} + 1\right) \leq 8(2L_\ell + 1) \leq 24L_\ell.$$

Combining these estimates, we obtain

$$\mathbb{E}(Y) \leq 10\sqrt{\ell L_\ell} + 24L_\ell \leq 24\left(\sqrt{\ell L_\ell} + L_\ell\right),$$

which proves Lemma 4.3. □

Now we are ready to derive the remaining lower bounds on $\pi(H_k^r)$ that were stated in the Introduction.

Proof of Theorem 1.2. Take any integers $3 \leq k \leq r+1$. Let $m := k-2$. Then $1 \leq m \leq r-1$. By using the circular construction, it is enough to show that the expectation of $\xi_{r,k-1}$ is at least the stated lower bound on $\pi(H_k^r)$.

We first suppose $m \leq r/2$. Since $Y_i^{(m)} = m + W_i^{(m)}$ for every i, taking the minimum over $1 \leq i \leq r$ gives $M_{r,m} = m + \min_{1\leq i\leq r} W_i^{(m)}$. Moreover, $\min_{1\leq i\leq r} W_i^{(m)} \geq -\max_{1\leq i\leq r} |W_i^{(m)}|$. It follows from Lemma 4.3 that

$$\mathbb{E}(M_{r,m}) \geq m - \mathbb{E}\left(\max_{1\leq i\leq r} |W_i^{(m)}|\right) \geq m - 24\left(\sqrt{m\log\frac{\mathrm{e}r}{m}} + \log\frac{\mathrm{e}r}{m}\right).$$

We now suppose $m > r/2$ and let $\ell = r - m$. Then $1 \leq \ell < r/2$. Since $m + \ell = r$, the terms in $Y_i^{(m)}$ and $Y_{i+m}^{(\ell)}$ together contain each of $X_1, \ldots, X_r$ exactly once. Therefore $Y_i^{(m)} + Y_{i+m}^{(\ell)} = S_r$ for every i. Since the map $i \mapsto i + m$ is a permutation of the indices modulo r, we obtain

$$M_{r,m} = S_r - \max_{1\leq i\leq r} Y_i^{(\ell)}.$$

Since $Y_i^{(\ell)} = \ell + W_i^{(\ell)}$ and $\mathbb{E}(S_r) = \sum_{j\leq r}\mathbb{E}(X_j) = r$, we have

$$\mathbb{E}(M_{r,m}) = r - \ell - \mathbb{E}\left(\max_{1\leq i\leq r} W_i^{(\ell)}\right) \geq m - \mathbb{E}\left(\max_{1\leq i\leq r} |W_i^{(\ell)}|\right).$$

By Lemma 4.3, we obtain

$$\mathbb{E}(M_{r,m}) \geq m - 24\left(\sqrt{\ell\log\frac{\mathrm{e}r}{\ell}} + \log\frac{\mathrm{e}r}{\ell}\right).$$

Recall that s was defined to be $\min\{k-2, r-k+2\} = \min\{m, r-m\}$ in the theorem. In both cases, we have

$$\mathbb{E}(M_{r,m}) \geq m - 24\left(\sqrt{s\log\frac{\mathrm{e}r}{s}} + \log\frac{\mathrm{e}r}{s}\right).$$

By the normalized exponential representation (Lemma 4.1) and Sidorenko's circular lower bound (Inequality (1.3)), we get

$$\pi(H_k^r) \geq \mathbb{E}(\xi_{r,m+1}) = \frac{1}{r}\,\mathbb{E}(M_{r,m}) \geq \frac{k-2}{r} - \frac{24}{r}\left(\sqrt{s\log\frac{\mathrm{e}r}{s}} + \log\frac{\mathrm{e}r}{s}\right).$$

This completes the proof of Theorem 1.2. □

Next, we prove Corollary 1.3, which illustrates the bound of Theorem 1.2 in some ranges of $k = k(r)$.

Proof of Corollary 1.3. Let $s := \min\{k-2, r-k+2\}$. By (1.1) and Theorem 1.2,

$$0 \leq \frac{k-2}{r} - \pi(H_k^r) \leq \frac{24}{r}\left(\sqrt{s\log\frac{\mathrm{e}r}{s}} + \log\frac{\mathrm{e}r}{s}\right). \tag{4.2}$$

We first assume that $k/\log r \to \infty$, which implies $\log(\mathrm{e}r/(k-2)) = o(k)$. If $k-2 \leq r/2$, then $s = k-2$ and

$$\sqrt{(k-2)\log\frac{\mathrm{e}r}{k-2}} + \log\frac{\mathrm{e}r}{k-2} = o(k).$$

If $k-2 > r/2$, then $s \leq r$ and $s \geq 1$, so

$$\sqrt{s\log\frac{\mathrm{e}r}{s}} + \log\frac{\mathrm{e}r}{s} \leq \sqrt{s\log(\mathrm{e}r)} + \log(\mathrm{e}r) \leq \sqrt{r\log(\mathrm{e}r)} + \log(\mathrm{e}r) = o(r).$$

Since $r < 2(k-2)$, the last estimate is also $o(k)$. Thus, in both cases, the right-hand side of (4.2) is $o(k/r)$, proving Part (i) of the corollary.

Finally, suppose that $k/r \to \alpha \in (0,1)$. Recall that $\beta = \min\{\alpha, 1-\alpha\}$. Then $s/r \to \beta$. Consequently,

$$\frac{24}{r}\left(\sqrt{s\log\frac{\mathrm{e}r}{s}} + \log\frac{\mathrm{e}r}{s}\right) = \frac{24}{\sqrt{r}}\sqrt{\frac{s}{r}\log\frac{\mathrm{e}r}{s}} + O\left(\frac{1}{r}\right) = \frac{24\sqrt{\beta\log(\mathrm{e}/\beta)} + o(1)}{\sqrt{r}},$$

as desired. □

Finally, we prove Corollary 1.4, which estimates the ratio between the upper bound in (1.1) and the new lower bounds as $r \to \infty$ for arbitrary $k = k(r)$. We will be rather brief and omit routine details.

Proof of Corollary 1.4. Let r be sufficiently large. By Corollary 1.3(i), it is enough to consider $k(r) = O(\log r)$ only.

Let G be the union of $b := \lfloor (k-1)/2 \rfloor$ independently and uniformly permuted copies of an H_3^r-free r-graph G' on $n \to \infty$ vertices of edge density $p = (\mathrm{e}^{-1} - o(1))/r$. Such an r-graph G' exists by Theorem 1.1, after deleting edges if necessary. Every $(r+1)$-set spans at most $2b \leq k-1$ edges in G and the expected edge density of G is

$$1 - (1-p)^b \geq bp - \binom{b}{2}p^2 \geq \left(\frac{1}{2\mathrm{e}} - o(1)\right)\frac{k-2}{r},$$

uniformly for $3 \leq k < O(\log r)$. This implies the first assertion.

For the second assertion, first note that $c_h \geq (1-o(1))h$ as $h \to \infty$ which can be seen by taking, for example, $\lambda := h - h^{2/3}$. Given $\varepsilon > 0$, fix an integer h sufficiently large and then $\eta > 0$ sufficiently small so that $(c_h - \eta)/(2h) > 1/2 - \varepsilon/2$. For all sufficiently large r, let G be the union of $b := \lfloor (k-1)/(2h) \rfloor$ randomly permuted copies of an H_{2h+1}^r-free r-graph of edge density

$p = (c_h - \eta)/r$, which exists by Theorem 1.1. Each $(r+1)$-set contains at most $2hb \le k-1$ edges in G and the expected edge density of G is at least

$$\frac{b(c_h-\eta)}{r} - O\left(\frac{b^2}{r^2}\right) = \left(\frac{c_h-\eta}{2h} - o(1)\right)\frac{k}{r} \ge \left(\frac{1}{2}-\varepsilon\right)\frac{k}{r},$$

implying the second assertion. □

5 Augmentations of t-graphs with zero kernel

In this section, we fix a non-empty t-graph F of uniformity $t \ge 2$. Recall that B_F denotes the edge–vertex incidence matrix of F and, for $r \ge t$, the r-augmentation $F(r)$ is the r-graph obtained from F by adding the same set of $r-t$ new vertices to every edge of F. We will refer to the set of the new $r-t$ vertices as the *stem* of $F(r)$. The vertices in $V(F)$ will be called *base* vertices.

Our proof of Theorem 1.6 will take some prime $q = q(r)$ depending on r (in order to define the first checksum in the finite field $\mathbb{F}_q$ with q elements). We will need q to satisfy certain properties. The following lemma lists these properties (with the third one being crucial in the proof of Theorem 1.6) and shows that there are only finitely many exceptional primes.

Lemma 5.1. *Suppose that B_F has zero kernel over $\mathbb{R}$. Then there is an integer Q_F such that, for every prime $q > Q_F$,*

(i) *t is non-zero in $\mathbb{F}_q$;*

(ii) *B_F has zero kernel over $\mathbb{F}_q$;*

(iii) *if $\boldsymbol{x} \in \mathbb{F}_q^{V(F)}$ and $B_F\boldsymbol{x} = d\mathbf{1}$ for some $d \in \mathbb{F}_q$, then $\boldsymbol{x} = (d/t)\mathbf{1}$.*

Proof. Let $h := |V(F)|$. Since B_F has rank h over $\mathbb{R}$, it has an $h \times h$ minor with non-zero determinant D. Since the matrix B_F is $\{0,1\}$-valued, D is an integer. Let Q_F be the largest prime divisor of $t|D|$. Let us verify that it satisfies the required properties. Take any prime $q > Q_F$. Then t and D are non-zero as elements of $\mathbb{F}_q$ and the above $h \times h$ minor remains nonsingular over $\mathbb{F}_q$, proving Items (i) and (ii). The last item is a consequence of the first two and the identity $B_F\mathbf{1} = t\mathbf{1}$: if $B_F\boldsymbol{x} = d\mathbf{1}$ then

$$B_F\left(\boldsymbol{x} - \frac{d}{t}\mathbf{1}\right) = d\mathbf{1} - \frac{d}{t}B_F\mathbf{1} = \mathbf{0}$$

and the zero kernel property over $\mathbb{F}_q$ gives $\boldsymbol{x} = (d/t)\mathbf{1}$, as desired. □

We also need a few auxiliary combinatorial lemmas. The following lemma shows that the zero kernel condition guarantees a positive Turán density for each augmentation.

Lemma 5.2. *If B_F has zero kernel over $\mathbb{R}$, then, for every $j \ge t$, the j-graph $F(j)$ is not j-partite and, consequently,*

$$\pi(F(j)) \ge \frac{j!}{j^j} > 0.$$

Proof. First, observe that the original t-graph F has no isolated vertices, since an isolated vertex would give a zero column of B_F. Also, F is not t-partite. Indeed, if $V(F) = V_1 \cup \cdots \cup V_t$ were a t-partition with every edge containing one vertex from every part, then the vector which is 1 on V_1, -1 on V_2, and 0 elsewhere would be a non-zero element of the kernel of B_F.

Suppose on the contrary that, for some $j \ge t$, the j-graph $F(j)$ is j-partite. Its $j-t$ stem vertices occur together in every edge and must therefore lie in distinct vertex classes. Every base vertex belongs to an edge, so the base vertices must lie inside the remaining t classes, and every base edge contains exactly one vertex from each of them. Thus F is t-partite. This contradiction proves the first part of the lemma.

For the second part, consider the complete balanced j-partite j-graph with $n \to \infty$ vertices. It is $F(j)$-free and its edge density is $j!/j^j + o(1)$, finishing the proof of the lemma. □

The standard averaging argument over vertex links can be applied to augmentations, giving the following monotonicity property.

Lemma 5.3. *For every $j \geq t$,*

$$\pi(F(j)) \geq \pi(F(j+1)).$$

Proof. Let $\mathcal{H}$ be an $F(j+1)$-free $(j+1)$-graph on n vertices with $\mathrm{ex}(n, F(j+1))$ edges. For each vertex v, its *link*

$$L_{\mathcal{H}}(v) := \left\{ A \in \tbinom{V(\mathcal{H})\setminus\{v\}}{j} : A \cup \{v\} \in \mathcal{H} \right\}$$

is $F(j)$-free: a copy of $F(j)$ in the link, with v adjoined to every edge, would give a copy of $F(j+1)$ in $\mathcal{H}$. Therefore

$$(j+1)|\mathcal{H}| = \sum_{v \in V(\mathcal{H})} |L_{\mathcal{H}}(v)| \leq n \,\mathrm{ex}(n-1, F(j)).$$

After division by $(j+1)\binom{n}{j+1}$, this becomes

$$\frac{\mathrm{ex}(n, F(j+1))}{\binom{n}{j+1}} = \frac{|\mathcal{H}|}{\binom{n}{j+1}} \leq \frac{\mathrm{ex}(n-1, F(j))}{\binom{n-1}{j}}.$$

The lemma follows by taking $n \to \infty$. □

The following well-known fact, proved by Katona, Nemetz and Simonovits [33], will also be used.

Lemma 5.4. *Let J be a j-graph. Then the ratio $\mathrm{ex}(n, J)/\binom{n}{j}$ is nonincreasing for $n \geq j$. In particular, $\mathrm{ex}(n, J)/\binom{n}{j} \geq \pi(J)$ for every $n \geq j$.*

Proof. For $n \geq j+1$, if $\mathcal{H}$ is an extremal J-free j-graph on n vertices, then the subgraph induced by a uniformly random $(n-1)$-subset has expected size $(n-j)|\mathcal{H}|/n$. Consequently,

$$\mathrm{ex}(n-1, J) \geq \frac{n-j}{n} \,\mathrm{ex}(n, J).$$

Since $\binom{n-1}{j} = \frac{n-j}{n}\binom{n}{j}$, division by $\binom{n-1}{j}$ proves the monotonicity. □

For a given j-graph J and integers m, n, the following simple probabilistic argument gives m J-free j-graphs on the common vertex set $[n]$ whose union is "large". We find it convenient to assume that the constructed j-graphs are edge-disjoint and encode them by a single map $\varphi : \binom{[n]}{j} \to \mathbb{Z}_m \cup \{\infty\}$. Recall that $\mathbb{Z}_m = \{0, \ldots, m-1\}$ denotes the set of residues modulo m. Here the j-graphs are the preimages of the elements of $\mathbb{Z}_m$ under φ while the set of uncovered edges is $\varphi^{-1}(\infty)$.

Lemma 5.5. *Let J be a fixed j-graph, let $0 < \beta < \pi(J)$, and let n, m be positive integers with $n \geq j$. Then there is a random map*

$$\varphi : \binom{[n]}{j} \to \mathbb{Z}_m \cup \{\infty\}$$

with all of the following properties.

(i) *For every outcome and every $b \in \mathbb{Z}_m$, the colour class $\varphi^{-1}(b)$ is J-free.*

(ii) *For every fixed $B \in \binom{[n]}{j}$,*

$$\mathbb{P}(\varphi(B) = \infty) \leq \mathrm{e}^{-\beta m}.$$

(iii) *The distribution of φ is invariant under every permutation of $[n]$.*

Proof. Take a maximum J-free j-graph $\mathcal{I}$ on $[n]$. By Lemma 5.4,

$$\frac{|\mathcal{I}|}{\binom{n}{j}} \geq \pi(J) > \beta.$$

For each $b \in \{0, \ldots, m-1\}$, choose a bijection

$$\psi_b : [n] \to [n]$$

uniformly at random, with the choices for different b being independent. For $B \in \binom{[n]}{j}$, let $\varphi(B)$ be the least $b \in \{0, \dots, m-1\}$ for which $\psi_b(B) \in \mathcal{I}$ and view b as an element of $\mathbb{Z}_m$. We define $\varphi(B) := \infty$ if no such b exists.

For fixed $B \in \binom{[n]}{j}$, the set $\psi_b(B)$ is a uniformly random j-subset of $[n]$. Each colour therefore succeeds in colouring B with probability $|\mathcal{I}|/\binom{n}{j} > \beta$. Hence, by independence,

$$\mathbb{P}(\varphi(B) = \infty) \le (1-\beta)^m \le \mathrm{e}^{-\beta m}.$$

For any finite colour $b \in \mathbb{Z}_m$, the colour-b class $\varphi^{-1}(b)$ is contained in

$$\psi_b^{-1}(\mathcal{I}) := \left\{ B \in \tbinom{[n]}{j} : \psi_b(B) \in \mathcal{I} \right\}.$$

This is just a relabelling of $\mathcal{I}$ so it is J-free.

Finally, precomposing all functions ψ_b with a fixed permutation of $[n]$ does not change their joint distribution. This establishes the permutation invariance of φ. □

The following result provides the key step of the proof.

Theorem 5.6. *Suppose that B_F has zero kernel over $\mathbb{R}$ and let Q_F be the integer returned by Lemma 5.1. Let integers $r > \ell \ge t$ satisfy $Q(r,\ell) \ge Q_F$, where*

$$Q(r,\ell) := \left\lceil \left(4 \binom{r}{\ell+1} \right)^{1/\ell} \right\rceil. \tag{5.1}$$

If $q > Q(r,\ell)$ is a prime then

$$\pi(F(r)) \ge \frac{3\pi(F(\ell))}{32q\log(2q)}. \tag{5.2}$$

Proof. Set

$$\beta := \frac{\pi(F(\ell))}{2} > 0 \quad \text{and} \quad m := \left\lceil \beta^{-1} \log(2q) \right\rceil.$$

Since $q \ge 2$ and $\beta \le 1/2$, the expression inside the ceiling is larger than 1. Therefore

$$m \le 2\beta^{-1}\log(2q) = \frac{4\log(2q)}{\pi(F(\ell))}. \tag{5.3}$$

Take any $n \ge r$. Let

$$a : [n] \to \mathbb{F}_q,$$

be a map chosen uniformly at random among all q^n choices. The *first checksum* of $A \in \binom{[n]}{r}$ is

$$\sigma_1(A) := \sum_{v \in A} a(v) \in \mathbb{F}_q.$$

The second checksum $\sigma_2(A)$ will be defined only for some r-sets A and we need to introduce some further random objects in addition to a in order to define it. For every $j \in \{t, \dots, \ell\}$, Lemma 5.3 gives $\pi(F(j)) \ge \pi(F(\ell)) > \beta$. Apply Lemma 5.5 to $J = F(j)$, with the same parameters β, n, m, and take the resulting random maps

$$\varphi_j : \binom{[n]}{j} \to \mathbb{Z}_m \cup \{\infty\}, \quad \text{for } j \in \{t, \dots, \ell\},$$

with the choices for different values of j being mutually independent, and also independent of a. For $A \subseteq [n]$ and $z \in \mathbb{F}_q$, define

$$A_z := \{v \in A : a(v) = z\}.$$

Thus A_z is the intersection of A with the fibre $a^{-1}(z)$.

We would like to define the *second checksum* of $A \in \binom{[n]}{r}$ as

$$\sigma_2(A) := \sum_{\substack{z \in \mathbb{F}_q \\ |A_z| \ge t}} \varphi_{|A_z|}(A_z).$$

In order for this expression to make sense, we need that

$$|A_z| \le \ell \text{ and if } |A_z| \ge t \text{ then } \varphi_{|A_z|}(A_z) \ne \infty, \quad \text{for every } z \in \mathbb{F}_q.$$

In this case, we say that $\sigma_2(A)$ *is defined* and view it as an element of $\mathbb{Z}_m$.

For each $\kappa_1 \in \mathbb{F}_q$ and $\kappa_2 \in \mathbb{Z}_m$, define an r-graph $\mathcal{F}_{\kappa_1,\kappa_2}$ on $[n]$ where an r-set $A \subseteq [n]$ is an edge of $\mathcal{F}_{\kappa_1,\kappa_2}$ if and only if

$$\sigma_1(A) = \kappa_1 \quad \text{and} \quad \sigma_2(A) = \kappa_2,$$

that is, the first checksum of A is κ_1 while the second checksum is defined and is equal to κ_2.

We first show that every realisation of $\mathcal{F}_{\kappa_1,\kappa_2}$ is $F(r)$-free. Suppose to the contrary that it contains a copy of $F(r)$. Let $D \subseteq [n]$ be the common stem of this copy, of size $r-t$, and identify its base vertices with $V(F)$.

For every $e \in E(F)$, the augmented edge $D \cup e$ has the first checksum κ_1, so

$$\sum_{u \in D} a(u) + \sum_{v \in e} a(v) = \kappa_1.$$

Thus all base-edge sums of the vector $(a(v))_{v \in V(F)}$ are equal to $d := \kappa_1 - \sum_{u \in D} a(u)$. Since $q > Q_F$, we have by Lemma 5.1(iii) that $a(v) = d/t$ for every $v \in V(F)$. Let

$$D' := \{u \in D : a(u) = d/t\} \quad \text{and} \quad t' := |D'| + t.$$

Note that, trivially, $t' \ge t$. For every augmented edge $A = D \cup e$, the set of its vertices v with $a(v) = d/t$ is exactly the t'-set $D' \cup e$, that is, $A_{d/t} = D' \cup e$. Since the second checksum of A is defined, we have that $t' \le \ell$. As e ranges over $E(F)$, the sets $D' \cup e$ form a copy of $F(t')$ in the fibre $a^{-1}(d/t) \subseteq [n]$.

All augmented edges A of the copy of $F(r)$ have the same occupancy profile $(|A_z|)_{z \in \mathbb{F}_q}$. In every fibre other than $a^{-1}(d/t)$, they contain exactly the same subset of the common stem D. Their second checksums are defined and are all equal to each other. By cancelling the contributions from the fibres other than $a^{-1}(d/t)$, we derive that there is $b \in \mathbb{Z}_m$ such that $\varphi_{t'}(D' \cup e) = b$ for every $e \in E(F)$. We have therefore obtained a copy of $F(t')$ in $\varphi_{t'}^{-1}(b)$ with $b \ne \infty$, contradicting Lemma 5.5. Thus $\mathcal{F}_{\kappa_1,\kappa_2}$ is $F(r)$-free.

Let us lower bound the expected edge density of $\mathcal{F}_{\kappa_1,\kappa_2}$ over uniform $(\kappa_1,\kappa_2) \in \mathbb{F}_q \times \mathbb{Z}_m$, independent of the previous choices. Fix an r-set A.

If some a-fibre contains at least $\ell+1$ vertices of A, then the function a is constant on some $(\ell+1)$-subset of A. For any fixed $(\ell+1)$-subset, this has probability $q^{-\ell}$. Hence, by the union bound,

$$\mathbb{P}\big(|A_z| > \ell \text{ for some } z \in \mathbb{F}_q\big) \le \binom{r}{\ell+1} q^{-\ell} \le \binom{r}{\ell+1} Q(r,\ell)^{-\ell} \le \frac{1}{4}.$$

Let us condition on a restriction of a to A such that each $|A_z|$ is at most ℓ. For each $z \in \mathbb{F}_q$ with $|A_z| \ge t$, the conditional probability of the bad event that $\varphi_{|A_z|}(A_z) = \infty$ is at most $\mathrm{e}^{-\beta m}$ by Lemma 5.5. By the union bound, the probability over the random functions $(\varphi_j)_{j=t}^{\ell}$ that there is $z \in \mathbb{F}_q$ with $|A_z| \ge t$ and $\varphi_{|A_z|}(A_z) = \infty$ is at most $q\mathrm{e}^{-\beta m} \le 1/2$.

Suppose that no bad event as above occurs. Then the second checksum of the r-set A is defined and assumes one of the m possible values in $\mathbb{Z}_m$. Since the first checksum $\sigma_1(A) \in \mathbb{F}_q$ is always defined, there is a choice of (κ_1,κ_2), namely $(\sigma_1(A),\sigma_2(A))$, such that $A \in \mathcal{F}_{\kappa_1,\kappa_2}$.

Putting everything together, the probability over the random choices of $(a,(\varphi_j),\kappa_1,\kappa_2)$ that a given r-set $A \subseteq [n]$ is an edge of $\mathcal{F}_{\kappa_1,\kappa_2}$ is at least

$$\left(1-\frac{1}{4}\right) \cdot \left(1-\frac{1}{2}\right) \cdot \frac{1}{q} \cdot \frac{1}{m} = \frac{3}{8qm}.$$

By the linearity of expectation, there is at least one outcome when $|\mathcal{F}_{\kappa_1,\kappa_2}| \ge \frac{3}{8mq}\binom{n}{r}$. Since this is true for every $n \ge r$ and every such realisation is $F(r)$-free, we conclude that

$$\pi(F(r)) \ge \frac{3}{8qm}.$$

Using (5.3), we obtain (5.2). □

Let us make some brief remarks about the above proof. The definition of the first checksum σ_1 and its usage (forcing all base vertices to have the same colour under the map a) are fairly standard, e.g. it appeared in the papers [35, 22, 49] that dealt with lower bounds on $\pi(H^r_{r+1})$. (These papers had a deterministic balanced colouring a and our choice of making the function a random was just a matter of convenience.) Here the zero-kernel condition ensures that equality of the first checksums forces all base vertices to have the same colour. The second checksum is more interesting and subtle. Informally, the main idea behind it is to extend the a-monochromatic base copy of F to the maximal monochromatic augmentation (with the stem being D' in the above proof) under the vertex colouring a and to explore the consequences of all edges of the original r-augmentation $F(r)$ having the same σ_2-value. A related use of sums of local colours and cancellation appears in a paper of Sales [47] on the Ramsey numbers of daisies.

The following intermediate result, which is a direct consequence of Theorem 5.6, will be useful in proving the lower bound of Theorem 1.6.

Lemma 5.7. *Suppose that B_F has zero kernel over $\mathbb{R}$. If r is sufficiently large, depending only on F, then it holds with $\ell := \lceil 8 \log r \rceil$ that*

$$\pi(F(r)) \geq \frac{3\pi(F(\ell))}{64r}. \tag{5.4}$$

Proof. Let $Q = Q(r, \ell)$ be defined by (5.1). The standard binomial estimate gives

$$\begin{aligned}\left(4\binom{r}{\ell+1}\right)^{1/\ell} &\leq \left(4\left(\frac{\mathrm{e}r}{\ell+1}\right)^{\ell+1}\right)^{1/\ell} \\ &= \frac{\mathrm{e}r}{\ell+1}\exp\left(\frac{\log 4 + \log(\mathrm{e}r/(\ell+1))}{\ell}\right).\end{aligned}$$

For all sufficiently large r,

$$\log 4 + \log\left(\frac{\mathrm{e}r}{\ell+1}\right) \leq 2\log r \leq \frac{\ell}{4}.$$

Consequently,

$$\left(4\binom{r}{\ell+1}\right)^{1/\ell} \leq \frac{\mathrm{e}^{5/4}r}{\ell+1} < \frac{4r}{\ell}.$$

Since $r/\ell \to \infty$, the ceiling contributes at most another r/ℓ for all sufficiently large r, and hence $Q \leq 5r/\ell$. Also $Q \to \infty$, so it eventually exceeds the constant Q_F returned by Lemma 5.1. Take a prime q with $Q < q < 2Q$; it exists by Bertrand's postulate. Then $q \leq 10r/\ell$. For large r, we have $2q \leq r$ and therefore

$$\log(2q) \leq \log r \leq \frac{\ell}{8}.$$

It follows that

$$q\log(2q) \leq \frac{10r}{\ell}\cdot\frac{\ell}{8} \leq 2r.$$

Theorem 5.6 now yields this lemma. □

Proof of Theorem 1.6. Fix $R = R(F)$ large enough so that, in particular, Lemma 5.7 applies for every $r \geq R$. Next, let r be sufficiently large. Define

$$r_1 := r \quad \text{and} \quad r_{i+1} := \lceil 8\log r_i \rceil \quad \text{for } i = 1, 2, 3.$$

We can assume that, for each $i \in [4]$, it holds that $r_i \geq R$. Applying (5.4) three times, we obtain

$$\pi(F(r)) \geq \frac{3\pi(F(r_2))}{64\,r_1} \geq \left(\frac{3}{64}\right)^2\frac{\pi(F(r_3))}{r_1 r_2} \geq \left(\frac{3}{64}\right)^3\frac{\pi(F(r_4))}{r_1 r_2 r_3}. \tag{5.5}$$

By Lemma 5.2 and the inequality $j! \geq (j/\mathrm{e})^j$, which holds for every integer $j \geq 1$, we have

$$\pi(F(r_4)) \geq \frac{r_4!}{r_4^{r_4}} \geq \mathrm{e}^{-r_4}. \tag{5.6}$$

Also, we can assume that $r_{i+1} \le 9 \log r_i$ for each $i \in [3]$. Thus $\mathrm{e}^{r_4} \le r_3^9 = (\log r)^{o(1)}$. Substituting this into (5.6) and then into (5.5), we conclude that

$$\pi(F(r)) \ge \left(\frac{3}{64}\right)^3 \frac{1}{r_1\, r_2\, r_3^{10}} \ge \frac{1}{r(\log r)^{1+o(1)}},$$

finishing the proof. □

Finally, Theorem 1.5 follows from Theorem 1.6 and the backward implication in the following easy result.

Lemma 5.8. *Let F be a connected graph with at least one edge. Then B_F has zero kernel if and only if F is non-bipartite.*

Proof. Suppose first that F is bipartite with parts X and Y. The non-zero vector $\boldsymbol{x}$ defined by

$$x_v := \begin{cases} 1, & \text{if } v \in X, \\ -1, & \text{if } v \in Y, \end{cases}$$

lies in the kernel of B_F, so B_F does not have zero kernel.

Conversely, suppose that F is non-bipartite and $B_F\boldsymbol{x} = 0$. For every edge uv,

$$x_u + x_v = 0. \tag{5.7}$$

Thus the values alternate in sign along any path. Since F contains an odd cycle, following (5.7) around that cycle gives $x_v = -x_v$ at a cycle vertex, and hence $x_v = 0$. Connectedness then propagates zero along paths to every vertex. Thus B_F has zero kernel. □

6 Concluding remarks

It is interesting that all the main results of this paper (including the analysis of the circular construction of Sidorenko) use probabilistic methods.

Of course, an obvious intriguing open problem is to determine the asymptotic value of $\pi(H_3^r)$ as $r \to \infty$.

In terms of r-augmentations of a fixed 3-chromatic graph F, (1.1) and Theorem 1.1 imply that $\pi(F(r)) = \Theta(1/r)$ if F has a triangle. In the next simplest case when F is the 5-cycle C_5, we have multiplicative gap of $(\log r)^{1+o(1)}$ between the bounds and closing it is an interesting open problem.

Open Problem 6.1. *Determine $\pi(C_5(r))$ within a constant factor as $r \to \infty$.*

A natural approach to improving the lower bound in Theorem 1.1 is to replace the cap set $\mathcal{C}_q$ by a set containing no k collinear points and take $h = 1$ in Lemma 3.4 (or to use some combination of both relaxing the condition on the set $\mathcal{C}_q$ while decreasing h appropriately). We do not know what general improvement can be achieved this way, so we briefly discuss just the case $k = 4$ (for which the multiplicative gap between the current asymptotic bounds is the largest). Here, the best lower bound coming from the results of this paper is

$$\pi(H_4^r) \ge \frac{\mathrm{e}^{-1} - o(1)}{r}, \tag{6.1}$$

the same as the bound for $\pi(H_3^r)$. Define $m(q)$ to be the maximum size of a set $\mathcal{X} \subseteq \mathrm{PG}(3, q)$ that contains no 4 collinear points. If we can improve the bound $m(q) \ge q^2 + 1$ coming from Lemma 3.2 by a factor $1 + \Omega(1)$ for a sequence of prime powers q whose consecutive terms have ratio $1 + o(1)$ then this will improve (6.1). Such an improvement is consistent with our current knowledge: Bramwell and Wilson [8] proved that $m(q) \le 2q^2$ for $q \ge 4$, while the best known general upper bound by Hill [31] improves it only in lower order terms. The best lower bound valid for every large prime q that the authors are aware of is $m(q) \ge (q+1)^2$ coming from a smooth cubic surface $\mathcal{X}$ of Frobenius class C_{12}, see Banwait, Fité and Loughran [5, Eq. (2.1) and Table 7.1] and Loughran and Trepalin [36, Theorem 1.1]. Such a surface has $(q+1)^2$ rational points and contains no line

defined over $\mathbb{F}_q$. Restricting its cubic equation to any line defined over $\mathbb{F}_q$ therefore gives at most three zeros.

In fact, if we had a set $\mathcal{X} \subseteq \mathrm{PG}(3,q)$ proving $m(q) \geq (1+\varepsilon)q^2$ for some fixed $\varepsilon > 0$ and infinitely many q, then by averaging over the planes of $\mathrm{PG}(3,q)$ we would get a plane section of $\mathcal{X}$ of size at least $(1+\varepsilon-o(1))q$, obviously still without any four points on a line. This would settle an open problem of Blokhuis for whose resolution he offered 10,000 Hungarian forints (see e.g. [4, Section 5, Problem 2]). Thus a constant factor improvement through larger generalised caps seems to require a substantial advance in the underlying finite geometry.

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Declaration

During the preparation of this work, the authors used AI systems (Chat GPT 5.6 and 6) for various tasks, including generation of candidate proofs. All AI suggestions that made into the final version were verified and streamlined by the authors (who also tried to find closest matches in the literature to the key AI-generated mathematical ideas used in this paper). The authors take full responsibility for the correctness of the paper.

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